

AccessWatch

Multimodal Fusion for Hierarchy-Aware Forecasting of Humanitarian
Access Risk in Ukraine

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Problem

Why this problem matters:

- Humanitarian access conditions can worsen rapidly in active conflict zones.
- Formal reporting often lags behind what is happening on the ground.
- Access risk is shaped by multiple factors, including conflict activity, infrastructure disruption, and geographic exposure.
- A system built from only one signal source may miss important warning patterns.

Motivation and Gap

What is missing in existing approaches:

- Many forecasting systems rely on randomly shuffled evaluation, which does not reflect real deployment conditions.
- Some reduce the task to a single binary outcome and miss different forms of escalation.
- Some assume all modalities are always available, even though real-world data can be incomplete or noisy.
- We wanted a system that is practical, interpretable, and robust under missing-modality settings.

Our goal:

Build a technically credible short-horizon early warning framework that is useful even under imperfect data availability.

Proposed Solution

Our solution: AccessWatch

- AccessWatch is a multimodal direct forecasting framework for Ukraine at the raion-week level.

Datasets

- ACLED conflict-event history
- NASA FIRMS thermal anomaly data
- VIIRS Black Marble nighttime-light data
- Population, settlement, transport, and border-access context
- Pre-2024 UNOSAT damage summaries

Outputs

- Any-event forecasting
- High-intensity forecasting
- Event-type forecasting
- Count prediction

Forecast horizons

- week+1 through week+5

Data and Experimental Setup

How we evaluated the system

- Forecasting panel built over 139 raions and 104 weekly anchors
- Strict chronological split:
 - 78 training weeks
 - 13 validation weeks
 - 13 test weeks
- Two feature sets:
 - Base multimodal input
 - Lagged-ACLEd-enhanced input
- Five model families:
 - Linear baseline
 - LightGBM
 - CatBoost
 - GRU
 - TCN
- Five missing-modality settings
- Total: 250 held-out test configurations

Why this matters:

This is not just a single-model demo; it is a structured benchmarking setup.

Best Balanced Result

Best balanced full-input system

Lagged-ACLED-enhanced + CatBoost

week+1 test performance

- Any-event F1: 0.8256
- High-intensity F1: 0.8643
- Event-type macro-F1: 0.7252

Across horizons

- Any-event F1 remains stable from 0.8172 to 0.8276
- High-intensity F1 stays above 0.85 through week+4
- Event-type macro-F1 remains in the 0.7133–0.7252 range

Takeaway:

The model remains strong even as forecast distance increases.

Main Findings

What we learned

- The strongest and most consistent gain came from adding recent conflict history.
- Under full input, lagged ACLED features improved:
 - Any-event F1 from 0.7947 to 0.8194
 - High-intensity F1 from 0.7748 to 0.8448
 - Event-type macro-F1 from 0.6537 to 0.6963
- More modalities did not automatically produce better forecasts.
- The value of FIRMS and nighttime-light inputs depended on the learner and the forecast horizon.

Main insight:

Effective multimodal forecasting is about signal complementarity, not simply adding more data.

Impact

Why this project is useful

- Provides short-horizon warning signals before conditions are fully visible in formal reporting
- Supports more structured forecasting than a simple yes/no warning
- Remains meaningful under incomplete modality availability
- Offers a practical path toward data-driven humanitarian early warning

Real-world value:

AccessWatch can help move decision-making from reactive reporting toward proactive monitoring and prioritization.

Limitations and Future Work

Current limitations

- The system forecasts access-related disruption signals, not direct humanitarian access ground truth
- Weekly aggregation and raion-level modeling may smooth short-lived local changes
- Satellite features are indirect proxies
- UNOSAT inputs are static pre-2024 summaries
- High-intensity is a rule-based target

Future work

- Add more direct access-ground-truth signals
- Incorporate more frequently updated damage and infrastructure layers
- Improve uncertainty calibration
- Explore finer spatiotemporal forecasting beyond the current raion-week setting

Demo / Closing

Demo

- Show forecast output or dashboard
- Highlight how predictions vary by region and horizon
- Show how the system remains usable when some modalities are removed

AccessWatch demonstrates that a rapidly developed research system can still deliver rigorous evaluation, meaningful empirical insight, and a credible path toward more operational humanitarian early warning tools.